

**The 6th International Conference on
Social Sciences and Intelligence Management (SSIM 2026)**
December 17–19, 2026, Chaoyang University of Technology, Taichung, Taiwan

**Special Session on
Computational Technologies
for Early Childhood Development and Interactive Learning
Call for Papers**

Organizers: Dr. Stephanie Fu Yang and Dr. Hua-Huei Chiou

Introduction

Intelligent technologies — including artificial intelligence, machine learning, multimodal sensing, educational robotics, and immersive computing — are rapidly transforming how young children’s development is assessed, supported, and enriched. These systems generate rich behavioral, physiological, and interaction data that, when processed through computational frameworks, offer unprecedented insights into cognitive growth, language acquisition, social-emotional development, and motor learning in early childhood.

This special session, “Intelligent Systems and Computational Technologies for Early Childhood Development and Interactive Learning,” invites original research that designs, implements, evaluates, or critically examines computational systems and data-driven methods applied to early childhood education and development (birth through age 8). We particularly encourage submissions presenting novel algorithms, system architectures, sensor-based frameworks, multimodal data pipelines, or AI-driven assessment tools grounded in rigorous empirical evaluation.

The session brings together researchers in computer science, educational technology, human–computer interaction, developmental informatics, and special education technology to advance intelligent, accessible, and developmentally appropriate technological solutions for young learners.

Topics of Interest

Topics include but are not limited to the following:

I. AI-Driven Developmental Assessment and Learning Analytics

- Machine learning models for automated developmental screening and early identification of delays
- AI-supported assessment of language acquisition, literacy emergence, and cognitive milestones
- Multimodal learning analytics integrating video, audio, physiological, and behavioral data streams
- Predictive modeling of developmental trajectories and learning outcome forecasting
- Intelligent dashboards for early childhood educators and clinicians: design, usability, and impact

II. Educational Robotics and Interactive Learning Environments

- Design and evaluation of child-robot interaction (CRI) systems for developmental support

- Social robots as learning companions: behavioral modeling, adaptive interaction, and affect recognition
- Intelligent interactive learning environments: architecture, sensor fusion, and real-time adaptation
- Gamification algorithms and adaptive game mechanics for early childhood learning systems
- Tangible computing and embodied interaction technologies for young learners

III. Immersive and Sensor-Based Technologies

- Augmented reality (AR) and virtual reality (VR) systems: design frameworks and developmental outcome evaluation
- Sensor-based movement analysis for motor development assessment and physical activity monitoring
- Wearable technologies and physiological sensing in early childhood contexts
- Computer vision and pose estimation for behavioral observation and developmental coding
- Digital developmental frameworks integrating multimodal sensor data for longitudinal tracking

IV. Computational Approaches to Inclusive and Special Education

- AI-assisted identification and support for children with autism spectrum disorder (ASD), developmental delays, or diverse learning profiles
- Assistive technology systems: design principles, computational architectures, and accessibility evaluation
- Technology-enhanced intervention delivery: adaptive algorithms and fidelity monitoring
- Natural language processing (NLP) for communication support and augmentative and alternative communication (AAC)
- Computational tools for therapists and special educators: intelligent decision-support systems

V. Music, Creative, and Social-Emotional Technology Systems

- Music and arts-based technology systems: computational design, interaction modeling, and developmental evaluation
- Affective computing and emotion recognition in early childhood learning contexts
- AI systems for supporting social-emotional learning (SEL): detection, modeling, and adaptive response
- Computational creativity tools: generative AI for children's creative and play-based learning
- Digital tools for language, literacy, and communication development: NLP-assisted scaffolding

Submission Guidelines

We invite original research papers, system demonstrations, empirical studies, and technical reports. Submissions should present novel computational, system-design, or data-driven contributions evaluated with appropriate empirical methods. Submissions that are purely descriptive, qualitative-only, or lack a technological or computational component are outside the scope of this track.

Submission requirements:

- Full papers: 12–15 pages (Springer LNCS/CCIS format), presenting complete research with system design, methodology, results, and evaluation
- Short papers: 6–11 pages, presenting work-in-progress, system prototypes, or pilot studies
- All submissions must follow the Springer LNCS/CCIS author template available on the SSIM 2026 website

- Papers must be submitted via the Microsoft CMT system, selecting Subject Area: Track 2 — Intelligent Systems and Computational Technologies for Early Childhood Development and Interactive Learning
- All submissions will undergo double-blind peer review by at least three qualified reviewers with expertise in educational technology, AI, HCI, developmental informatics, or special education technology

Please visit the SSIM 2026 official website for templates and submission instructions:

<https://ssim2026.cyut.edu.tw>

Publication

Papers that are presented orally at the conference and meet the publication requirements of SSIM 2026 conference proceedings will be considered for publication in Springer conference proceedings. The primary publication plan is within Springer's Communications in Computer and Information Science (CCIS) series. Depending on the thematic focus of accepted papers and the final publication arrangement, selected papers may also be included in a Springer Information and Computer Science / Intelligent Systems-related proceedings series, similar to the publication model adopted by SSIM 2025.

All publication arrangements are subject to Springer's final review and approval processes. Accepted papers must satisfy the academic quality standards and formatting requirements specified by both SSIM 2026 and the designated Springer publication series.

Conference proceedings published by Springer are typically submitted for indexing in major academic databases, which may include Scopus, depending on the specific series and indexing policies at the time of publication.

Important Dates

- **Abstract Submission Deadline:** September 10, 2026
- **Notification of Abstract Acceptance:** September 25, 2026
- **Full Paper Submission Deadline:** October 15, 2026
- **Notification of Full Paper Acceptance:** November 20, 2026
- **Camera-Ready Paper Deadline:** December 10, 2026
- **Conference Dates:** December 17–19, 2026

If SSIM extends the dates, updated deadlines will be announced on the official conference website.

Contact

For further information, please contact the session organizers:

Dr. Hua-Huei Chiou

Department of Early Childhood Development and Education,
Chaoyang University of Technology, Taichung, Taiwan

Email: hhchiou@gm.cyut.edu.tw

Dr. Stephanie Fu Yang

Department of Psychology,
National Cheng Kung University

Email: yangsf@gs.ncku.edu.tw